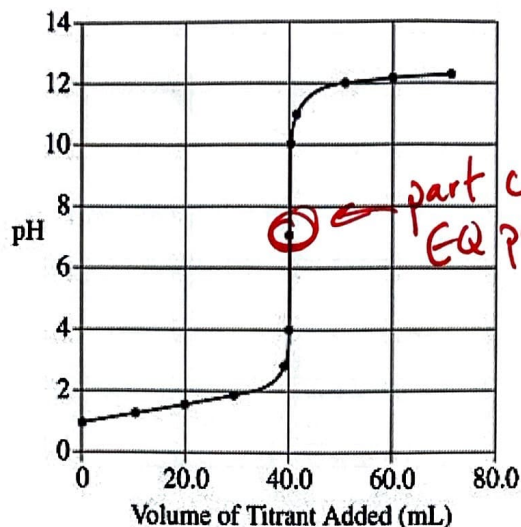




1. A solution of 0.100 M HCl and a solution of 0.100 M NaOH are prepared. A 40.0 mL sample of one of the solutions is added to a beaker and then titrated with the other solution. A pH electrode is used to obtain the data that are plotted in the titration curve shown above.

- a. Write the net ionic equation for the reaction of HCl(aq) and NaOH(aq). (1 point)



- b. Identify the solution that was initially added to the beaker. Explain your reasoning. (1 point)

HCl b/c pH was 1 @ initially, matching the 0.1M soln of H^+

- c. On the titration curve above, circle the point that corresponds to the equivalence point. (1 point)

- d. At the equivalence point, how many moles of titrant have been added? (1 point)

$$\text{moles titrant} = \text{moles base} = 40.0 \text{ mL} \cdot \frac{1}{1000} \cdot 0.1 \text{ M} = 0.00400 \text{ mol}$$

- e. The same titration is to be performed again, this time using an indicator. Use the information in the table below to select the best indicator for the titration. Explain your choice. (2 points)

Indicator	pH Range of Color Change
Methyl violet	0 – 1.6
Methyl red	4 – 6
Alizarin yellow	10 – 12

Use methyl red.

The curve shows that the equivalence point occurs over pH 4–10, and this is within the range of the M.R.

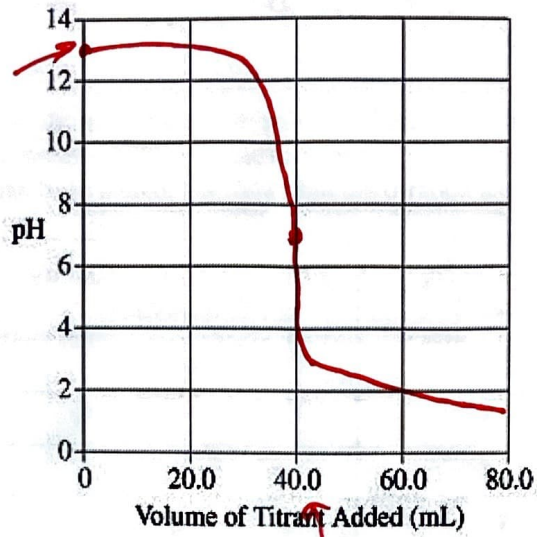
- f. What is the difference between the equivalence point of a titration and the end point of a titration? (1 point)

- g. On the grid provided on the next page, sketch the titration curve that would result if the solutions in the beaker and buret were reversed (i.e., if 40.0 mL of the solution used in the buret in the previous titration were titrated with the solution that was in the beaker). (3 points)

f. EQ point = where moles H^+ = moles OH^- from the titrant & analyte.

The endpoint = where the indicator changes color. They should be very similar for a well selected indicator.

g.)
0.1M
NaOH
pH=13



Eq
still pH 7 @ 40 mL

T2

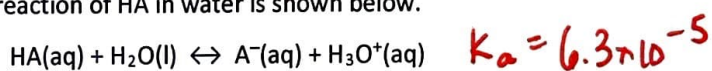
2. A 1.22 g sample of a pure monoprotic acid, HA, was dissolved in distilled water. The HA solution was then titrated with 0.250 M NaOH. The pH was measured throughout the titration, and the equivalence point was reached when 40.0 mL of the NaOH solution had been added. The data from the titration are recorded in the table below.

Volume of 0.250 M NaOH Added (mL)	pH of Titrated Solution
0.00	?
10.0	3.72
20.0	4.20
30.0	?
40.0	8.62
50.0	12.40

← EQ pt.

- Explain how the data in the table above provide evidence that HA is a weak acid rather than a strong acid. (1 point)
- Write the balanced net-ionic equation for the reaction that occurs when the solution of NaOH is added to the solution of HA. (1 point)
- Calculate the number of moles of HA that were titrated. (1 point)
- Calculate the molar mass of HA. (1 point)

The equation for the dissociation reaction of HA in water is shown below.



- Assume that the initial concentration of the HA solution (before any NaOH solution was added) is 0.200 M. Determine the pH of the initial HA solution. (3 points)

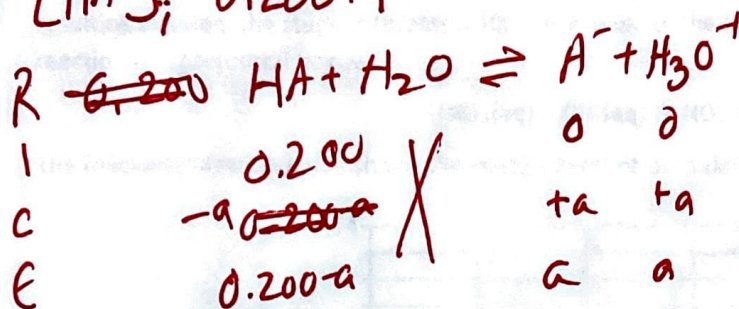
a.) HA is a weak acid because at the equivalence point, only the conjugate base, A^- , is contributing to the pH. Because the pH is basic (>7) the conjugate base must be a significant species, indicating it came from a weak acid.



c.) Moles A = moles B (1:1 mol/mol ratio) = 0.250 M NaOH • 0.0400 L ~~10.0~~
 mol A = 0.0100 moles HA

$$d.) \text{MM} = \frac{\text{mass}}{\text{moles}} = \frac{1.22 \text{ g}}{0.0100 \text{ moles}} = 122 \text{ g/mol}$$

$$e.) [\text{HA}]_i = 0.200 \text{ M}$$



$$K_a = \frac{[\text{A}^-][\text{H}_3\text{O}^+]}{[\text{HA}]}$$

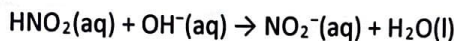
$$\cancel{2.1 \times 10^{-5}} = 6.3 \times 10^{-5} = \frac{a^2}{(0.200-a)} \quad a \text{ is small!}$$

$$\text{pH} = -\log [\text{H}_3\text{O}^+] = -\log (3.55 \times 10^{-3}) = \underline{\underline{2.45}}$$

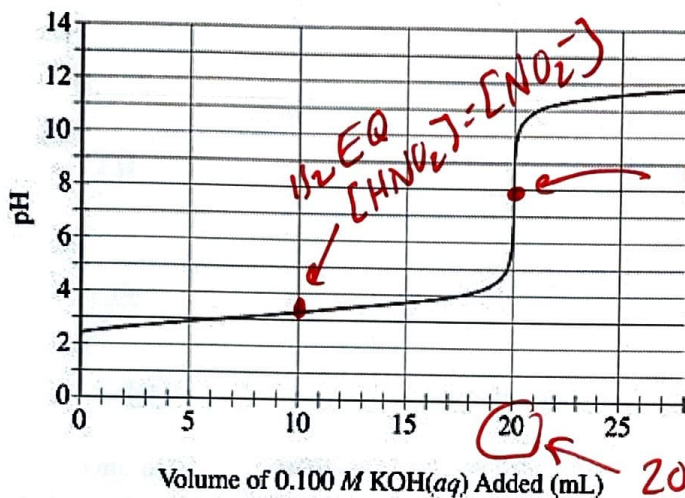
$$a = 3.55 \times 10^{-3} = [\text{H}^+]$$

T3

3. To produce an aqueous solution of HNO_2 , a student bubbles $\text{N}_2\text{O}_3(\text{g})$ into distilled water. Assume that the reaction goes to completion and that HNO_2 is the only species produced. To determine the concentration of $\text{HNO}_2(\text{aq})$ in the resulting solution, the student titrates a 100. mL sample of the solution with 0.100 M $\text{KOH}(\text{aq})$. The neutralization reaction is represented below.



The following titration curve shows the change in pH of the solution during the titration.



- Use the titration curve and the information above to
 - determine the initial concentration of the $\text{HNO}_2(\text{aq})$ solution (1 point)
 - estimate the value of the pK_a for $\text{HNO}_2(\text{aq})$ (1 point)
- During the titration, after a volume of 15 mL of 0.100 M $\text{KOH}(\text{aq})$ has been added, which species, $\text{HNO}_2(\text{aq})$ or $\text{NO}_2^-(\text{aq})$, is present at a higher concentration in the solution? Justify your answer. (1 point)



$$\text{mol A} = \text{mol B}$$

$$M_A V_A = M_B V_B$$

$$M_A \cdot 100 = 0.100 \cdot 20$$

$$M_A = 0.0200 \text{ M HNO}_2$$

ii) $\text{pK}_a = \text{pH} @ \frac{1}{2} \text{EQ}$
 (10 mL)
 $= 3.7$

b.) ~~when~~ NO_2^- has a higher concentration than HNO_2
 @ 15 mL, over $\frac{1}{2}$ of the acid is neutralized into

NO_2^- . $\frac{15}{20}$ or $\frac{3}{4}$ of the HA is gone & $\frac{1}{4}$ remains